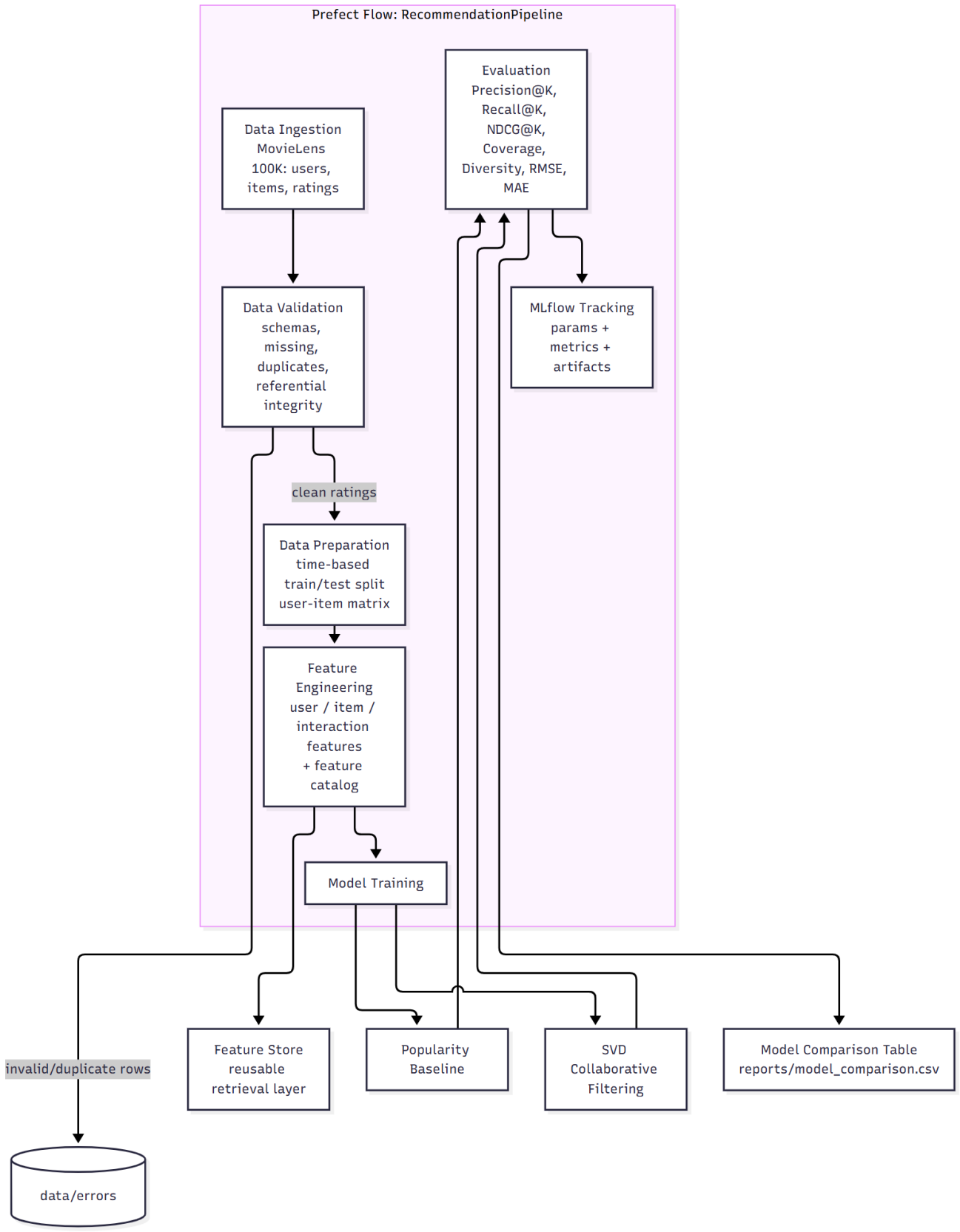
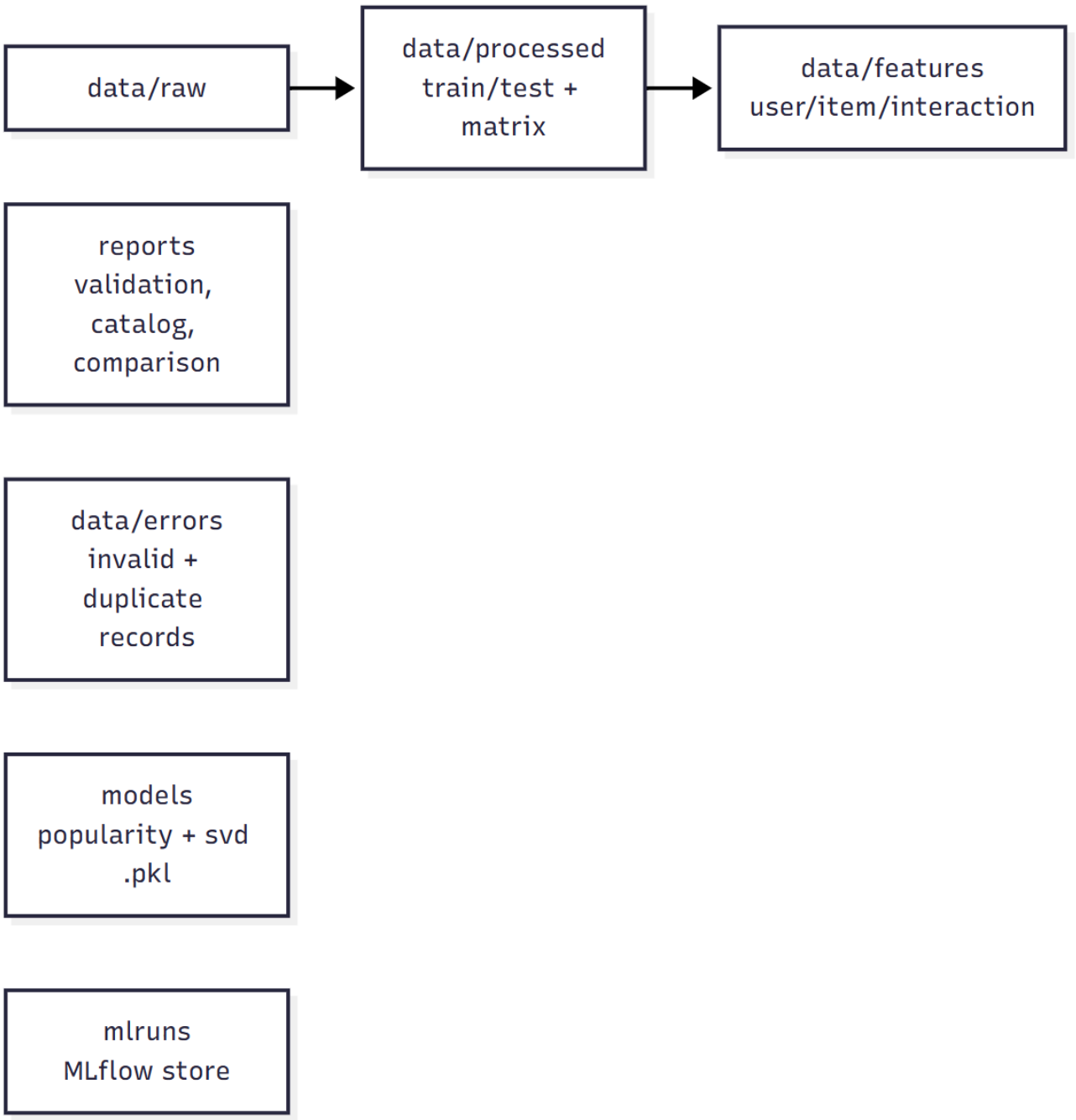
Pipeline Architecture

High-Level Flow



Storage Layout



Components

Layer	Module	Responsibility
Config	`src/utils.py` + `config/config.yaml`	Central config + path helpers
Ingestion	`src/download_dataset.py` , `src/ingestion.py`	Download + load users/items/ratings (incl. genres)
Validation	`src/validation.py`	Schema + integrity checks; separate error records; return clean data
Preparation	`src/preparation.py`	Time-based split, user-item matrix

Layer	Module	Responsibility
Features	`src/feature_engineering.py`, `src/feature_store.py`	Feature creation, catalog, retrieval layer
Models	`src/popularity_model.py`, `src/collaborative_model.py`	Baseline + SVD
Evaluation	`src/evaluation.py`	Ranking + rating metrics, comparison table
Tracking	`src/mlflow_tracking.py`	MLflow params/metrics/artifacts
Orchestration	`orchestration/prefect_pipeline.py`	Prefect flow wiring all stages